

$$1. (a) g_m = \sqrt{2 I_D \mu_n C_{ox} \left(\frac{W}{L}\right)_1} = 5.182 \text{ mS}$$

$$A_v = -g_m R_D = -10.364$$

$$(b) I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_1 V_{ov}^2$$

$$V_{out} = V_{DD} - I_D R_D$$

$$V_{ov} = V_{in} - V_{TH}$$

$$A_v = -g_m (R_D)$$

$$\text{可得 } V_{in} = 1.137 \text{ V} \quad A_v = -11.73$$

$$(c) V_{in} - V_{TH} - V_{out} = 50 \times 10^{-3}$$

$$V_{out} = V_{DD} - I_D R_D$$

$$I_D = \mu_n C_{ox} \left(\frac{W}{L}\right)_1 \left((V_{in} - V_{TH}) V_{out} - \frac{V_{out}^2}{2} \right)$$

$$A_v = -g_m R_D$$

$$\text{可得 } A_v = -4.51$$

2.

$$(1) \text{ 对于 } M_2, V_{SB} = V_{out}$$

$$V_{THN2} = V_{TH0} + \gamma (\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$$

$$I_{D2} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_2 (V_{DD} - V_{out} - V_{THN2})^2 = 0.5 \times 10^{-3}$$

$$\text{可得 } V_{out} \approx 1.466 \text{ V}$$

$$g_{m1} = \sqrt{2 I_{D1} \mu_n C_{ox} \left(\frac{W}{L}\right)_1} = 3.66 \text{ mS}$$

$$g_{m2} = \sqrt{2 I_{D2} \mu_n C_{ox} \left(\frac{W}{L}\right)_2} = 1.64 \text{ mS}$$

$$g_{mb2} = g_{m2} \frac{\gamma}{2\sqrt{2\phi_F + V_{SB}}} = 0.204 \text{ mS}$$

$$r_{o1} = r_{o2} = \frac{1}{\lambda I_{D1}} = 20 \text{ k}\Omega$$

$$A_v = -g_{m1} \left(\frac{1}{g_{m2} + g_{mb2}} \parallel r_{o2} \parallel r_{o1} \right) \approx -1.886$$

$$(2) A_v = -g_{m1} \cdot \left(\frac{1}{g_{m2}} \parallel r_{o2} \parallel r_{o1} \right)$$

$$\text{其中 } g_{m1} = 3.66 \text{ mS}$$

$$g_{m2} = 1.64 \text{ mS}$$

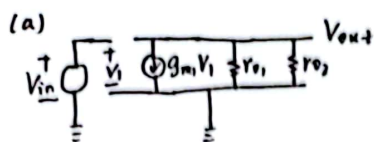
$$r_{o1} = \frac{1}{\lambda_n I_{D1}} = 20 \text{ k}\Omega$$

$$r_{o2} = \frac{1}{\lambda_p I_{D2}} = 10 \text{ k}\Omega$$

$$\text{故 } A_v \approx -3.5716.$$



3.



(b) $V_1 = V_{in}$

$$V_{out} = -g_{m1} V_{in} (r_{01} \parallel r_{02})$$

↓

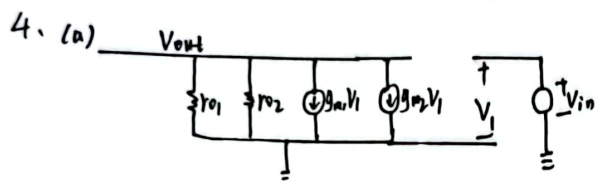
$$\frac{V_{out}}{V_{in}} = -g_{m1} (r_{01} \parallel r_{02})$$

$$r_{out} = r_{01} \parallel r_{02}$$

(c) $V_{DD} - V_{out} \geq V_{DD} - V_b - |V_{THP}| \Rightarrow V_{out} \leq V_b + |V_{THP}|$

同时 $V_{out} \geq V_{in} - V_{THN}$

综合上 $V_{in} - V_{THN} \leq V_{out} \leq V_b + |V_{THP}|$

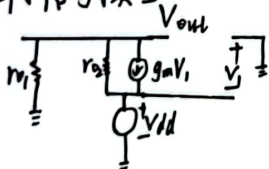


$$V_{out} = -(g_{m1} + g_{m2}) V_{in} \cdot (r_{01} \parallel r_{02})$$

$$A_v = \frac{V_{out}}{V_{in}} = -(g_{m1} + g_{m2}) \cdot (r_{01} \parallel r_{02})$$

(b) 先求 A_{vdd}

画小信号模型

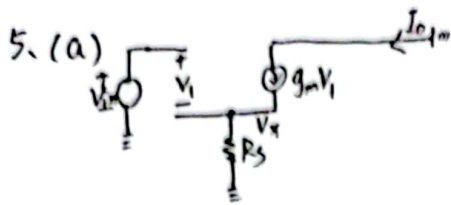


$$\frac{V_{out}}{r_{01}} = \frac{V_{dd} - V_{out}}{r_{02}} + g_m V_{dd}$$

$$\text{可得 } A_{vdd} = \frac{V_{out}}{V_{dd}} = \frac{r_{01} (1 + g_m r_{02})}{r_{01} + r_{02}}$$

$$\begin{aligned} PSRR &= \frac{A_v}{A_{vdd}} = -(g_{m1} + g_{m2}) \cdot (r_{01} \parallel r_{02}) \cdot \frac{r_{01} + r_{02}}{r_{01} (1 + g_m r_{02})} \\ &= \frac{-(g_{m1} + g_{m2}) r_{02}}{1 + g_m r_{02}} \end{aligned}$$





$$\begin{cases} I_D = \frac{V_{gs}}{R_S} \\ V_i = V_{in} - V_{gs} \Rightarrow I_D = \frac{g_m}{1+g_m R_S} V_{in} \quad \text{故 } G_m = \frac{g_m}{1+g_m R_S} \\ g_m V_{gs} = I_D \end{cases}$$

(b) $g_m = \sqrt{2I_D \mu_n C_{ox} \frac{W}{L}} = 3.66 \text{ mS}$

$$G_m = \frac{g_m}{1+g_m R_S} = \frac{3.66 \times 10^{-3}}{1+3.66 \times 10^{-3} \times 200} = 2.11 \text{ mS}$$

$$A_v = -G_m R_D = -4.22$$

(c)

$$\begin{cases} I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{gs1} - V_{TH})^2 \quad ① \\ V_{in} - V_{gs1} - V_{TH} = V_{out} - V_{gs1} \quad ② \\ V_{gs1} = I_D R_S \quad ③ \\ V_{out} = V_{DD} - I_D R_D \quad ④ \end{cases}$$

将②、③、④代入①可得

$$\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{DD} - I_D (R_D + R_S))^2 = I_D$$

$$g_m = \sqrt{2I_D \mu_n C_{ox} \frac{W}{L}}$$

$$A_v = -\frac{g_m}{1+g_m R_S} R_D = -5.297$$

6.

(a) $\frac{V_o}{V_i} = g_m \cdot \frac{1}{g_m} = 1$

(b) $\because V_{DD} - V_o > V_i - V_o - V_{THN}$, 故晶体管始终饱和

~~$$\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (3 - V_o - V_{THN})^2 = I_D = 200 \mu\text{A}$$~~

有 $\begin{cases} I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (3 - V_o - V_{THN})^2 = 200 \mu\text{A} \\ V_{THN} = 0.7 + \gamma(\sqrt{2\phi_F + V_o} - \sqrt{2\phi_F}) \end{cases}$

可解出 $V_o = 1.49 \text{ V}$

$$A_v = g_m \cdot \frac{1}{g_m + g_{mb}} = \frac{1}{1+\eta}, \quad \eta = \frac{\gamma}{2\sqrt{2\phi_F + V_o}}$$

可得 $A_v \approx 0.93$

(c) $\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (3 - V_o - V_{THN})^2 = 200 \times 10^{-6} + \frac{V_o}{R}$

$$\begin{cases} V_{THN} = 0.7 + \gamma(\sqrt{2\phi_F + V_o} - \sqrt{2\phi_F}) \end{cases}$$

我们认为 V_{THN} 与 (b) 相同, 可得 $V_o = 1.48 \text{ V}$

$$A_v = g_m \cdot \left(\frac{1}{g_m + g_{mb}} \parallel R_o \right) = 0.92$$

(d) 与 (c) 同理, $A_v = 0.93$



$$7. \textcircled{1} R_i = \frac{r_o + R_D}{1 + g_m r_o}$$

$$r_o = \frac{1}{\lambda I_D} = 1 \text{ M}\Omega$$

$$g_m = \sqrt{2 I_D \mu_n C_{ox} \frac{W}{L}} = 2000 \mu\text{A/V}$$

$$\text{故 } R_i = \frac{1 \text{ M}\Omega + 1 \text{ k}\Omega}{2001} \approx 500 \Omega$$

$$\textcircled{2} R_i = \frac{1 \text{ M}\Omega + 1 \text{ M}\Omega}{2001} \approx 1000 \Omega$$

8.

$$(a) V_{in} - V_{THN0} = \sqrt{\frac{2 I_{D1}}{\mu_n C_{ox} (\frac{W}{L})_1}} = 0.273 \text{ V}$$

$$V_x = V_{in} - V_{THN0} + 0.05 = 0.323 \text{ V}$$

$$V_b - V_x - V_{THN0} = \sqrt{\frac{2 I_{D2}}{\mu_n C_{ox} (\frac{W}{L})_2}} = 0.61 \text{ V}$$

$$\text{故 } V_b = 1.633 \text{ V}$$

$$(b) g_{m1} = \sqrt{2 I_{D1} \mu_n C_{ox} (\frac{W}{L})_1} = 3.664 \text{ mS}$$

$$A_v = g_{m1} (R_D \parallel g_{m2} r_{o2} r_{o1}) \approx -3.644$$

(c) 我们先假设 M_1 先进线性区

则当 M_1 处在线性区边缘时有 $V_x = V_{in} - V_{THN0}$

$$I = \frac{1}{2} \mu_n C_{ox} (\frac{W}{L})_1 (V_{in} - V_{THN0})^2 \quad \textcircled{1}$$

$$I = \frac{1}{2} \mu_n C_{ox} (\frac{W}{L})_2 (V_b - V_x - V_{THN0})^2 \quad \textcircled{2}$$

$$V_{out} = V_{DD} - I R_D \quad \textcircled{3}$$

取 $V_b = 1.633 \text{ V}$ 可得: $V_x = 0.288 \text{ V}$ $V_{in} = 0.988 \text{ V}$ (联立①②得)

$$I = 0.557 \text{ mA}$$

$$V_{out} = V_{DD} - I R_D = 2.443 \text{ V}$$

此时 $V_{out} - V_x = 2.155 \text{ V} > V_b - V_x - V_{THN0} = 0.645 \text{ V}$, M_2 饱和

故此情况成立

综上可知, M_1 先进入线性区

当 $V_{in} = V_{THN0}$ 时, $I = 0$, $V_{out} = 3 \text{ V}$

所以我们可得 V_{out} 摆幅为 2.443 V $V_{out} \leq 3 \text{ V}$

(d) $V_{out} = V_{out_min}$ 时, 由(1)知, $V_x = 0.288 \text{ V}$

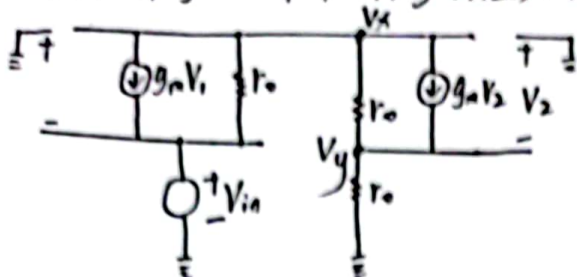
$$V_{out} = 3 \text{ V} \text{ 时, } V_x = V_b - V_{THN0} = 1.633 - 0.7 = 0.933 \text{ V}$$



9. (a) $R_{01} = r_o$ $R_{02} = r_o$ $R_{03} = r_o$

$R_{04} = g_m r_o^2$ $R_{05} = \frac{g_m r_o^2}{2}$

对于 R_{in} , 可以画出小信号模型如下:



可列出方程如下:

$$\begin{cases} g_m V_{in} + \frac{V_{in} - V_x}{r_o} = \frac{V_y}{r_o} \\ \frac{V_x - V_y}{r_o} - g_m V_y = \frac{V_y}{r_o} \\ I_{in} = \frac{V_y}{r_o} \end{cases}$$

最终可得 $\frac{V_{in}}{I_{in}} = \frac{(3 + g_m r_o) r_o}{1 + g_m r_o}$

又 $r_o \gg \frac{1}{g_m}$, 故 $R_{in} \approx \frac{g_m r_o^2}{g_m r_o} = r_o$ (也可以由 $R_{in} = \frac{r_o + R_D}{1 + g_m r_o}$ 直接求得, 见第三章 PPT 78 页)

(b) $R_o = (g_m r_o \cdot \frac{r_o}{2}) // g_m r_o^2 = \frac{1}{2} g_m r_o^2$

(c) 当输出接 AC 地时, 我们知道从 M_3 漏极往下的电阻 $R_D = g_m r_o^2 // 0 = 0$

故此时 $R_{in} = \frac{1}{g_m}$ (由 $R_{in} = \frac{r_o + R_D}{1 + g_m r_o}$ 计算)

而 $R_{02} = r_o \gg R_{in}$

当 V_i 变化 ΔV_i , 产生 $g_m \Delta V_i$ 电流, 这股电流会分流至 M_2 所在支路及 M_3 所在支路

$\therefore R_{in} \ll R_{02}$

$\therefore$ 主要电流会流入 M_3 所在支路, 故可以认为 $G_m \approx g_m$

(d) $\frac{V_o}{V_i} = -g_m \cdot R_o = -\frac{1}{3} g_m^2 r_o^2 = -\frac{1}{3} \times 4 \times 10^{-6} \times 15^2 \times 10^6 = -300$



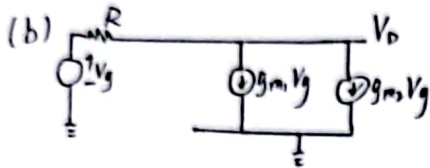
10.

(a) $\because R=1\text{M}\Omega$, 所以我们可以认为 G、D 之间无直流电流故 $V_G = V_D$

$$I_{D1} = I_{D2} = \frac{1}{2} \mu_p \left(\frac{W}{L} \right)_1 (V_{DD} - V_D - |V_{THP}|)^2$$

即由 $\mu_p \left(\frac{W}{L} \right)_1 = \mu_n \left(\frac{W}{L} \right)_2$ 、 $|V_{THP}| = |V_{THN}|$ 可知, 若想电流通过 M_1 、 M_2 相同, 需令 $V_{DD} - V_D = V_D$, 可得 $V_D = 1\text{V}$

$$\text{故 } I_{D1} = I_{D2} = 0.125\text{mA}$$

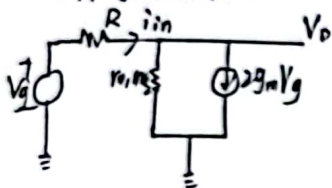


$$g_{m1} = g_{m2} = \sqrt{2I_D \mu_n \left(\frac{W}{L} \right)_1} = 0.5\text{mS}$$

$$V_D = V_G + (-2g_m V_G) \cdot R$$

$$\text{故 } \frac{V_D}{V_G} = 1 - 2g_m R = -999$$

(c) 小信号模型如下:



根据基尔霍夫定理可得:

$$\frac{V_D - V_G}{R} + \frac{V_D}{r_{o1} || r_{o2}} = -2g_m V_G \quad (1)$$

$$\text{其中 } r_{o1} || r_{o2} = \frac{r_o}{2}$$

$$r_o = \frac{1}{\lambda I_D} = 160\text{k}\Omega$$

由 (1) 可得

$$\frac{V_D}{V_G} = \frac{1 - 2g_m R}{1 + \frac{2R}{r_o}} = -74$$

$$\left\{ \begin{aligned} r_{in} &= \frac{V_G}{i_{in}} \end{aligned} \right.$$

$$i_{in} = 2g_m V_G + \frac{V_D}{\frac{r_o}{2}}$$

$$\text{故 } r_{in} = \frac{V_G}{2g_m V_G + \frac{2V_D}{r_o}} = \frac{1}{2g_m + \frac{2}{r_o} \cdot \frac{V_D}{V_G}} = 13.3\text{k}\Omega$$

$$(d) \frac{V_d}{V_{sig}} = \frac{V_d}{V_{in}} \cdot \frac{V_{in}}{V_{sig}} = -74 \cdot \frac{R_{in}}{R_{in} + R_1} = -74 \cdot \frac{13.3}{13.3 + 20} = -29.6$$

$$(e) \left\{ \begin{aligned} V_{DD} - V_D &\gg V_{DD} - V_G - |V_{THP}| \\ V_D &\gg V_G - V_{THN} \end{aligned} \right.$$

$$0.5\text{V} \leq V_D \leq 1.5\text{V}$$

